Personal Statement

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UK Global Talent Visa — Digital Technology (Exceptional Promise)

I am applying under the Global Talent route (Exceptional Promise) as a security engineer and technical founder working at the intersection of cybersecurity, machine learning, and privacy-preserving verification. My case is independently supported by verifiable outputs: a publicly deployed compliance platform, a sole-authored full-stack codebase, 38 Solidity files spanning core contracts, interfaces, and supporting modules, reproducible ML benchmarks, a live enterprise dashboard, security audit artefacts, four research preprints with documented academic uptake, and a UK patent filing. Together, these establish original innovation and a credible trajectory toward leadership in the UK's digital technology sector.

Strategic Motivation and the UK Roadmap

Britain leads globally in regulatory innovation. The **FCA's** evolving digital-asset framework, HM Treasury's stablecoin proposals, and the Bank of England's wholesale CBDC program have together increased demand for the exact type of infrastructure I have built **crypto-compliance middleware** that enables checks before a transaction goes through, while preserving privacy on-chain. AMTTP is designed as compliance-oriented middleware that helps regulated institutions apply Travel Rule workflows, meet risk-based monitoring requirements, and enforce policy rules directly at the protocol layer.

Over the next five years, I plan to establish a UK-headquartered RegTech company in **Derby**, with plans to recruit engineering and compliance talent within the UK ecosystem and to engage directly with London's regulatory and fintech community. Within that horizon I plan to **create 8–12 skilled engineering and compliance roles in the UK** (smart-contract, ML, zero-knowledge, and regulatory engineering) and convert AMTTP from a validated open protocol into a licensed **Compliance-as-a-Service** provider, supplying **crypto-compliance infrastructure** to UK fintechs, banks, and regulated digital-finance institutions. This aligns with the UK's broader focus on secure digital-finance innovation.

Innovation: The AMTTP Protocol

To address fragmentation in digital-asset compliance, I spent three years developing a modular compliance infrastructure framework as an independent technical founder. This development cycle ensured full ownership of the underlying intellectual property and system architecture before broader public deployment and commercial development. During this period, I engineered the core mathematical risk engine, the smart-contract enforcement layer, and the supporting security architecture.

Over the past eight months, I began publicly releasing elements of the work through open-source code, research preprints, and technical artefacts. This phase culminated recently in the live deployment of the Anti-Money Laundering Transaction Transfer Protocol (AMTTP), including a publicly accessible infrastructure environment, security-testing evidence, formal UK patent filings, and supporting research publications.

The result is the **Anti-Money Laundering Transaction Transfer Protocol (AMTTP)**. Unlike conventional tooling that acts after settlement, AMTTP enables risk decisions before a transaction is completed. As an **independent founder**, working without institutional sponsorship, venture funding, or a delivery team, I architected and delivered the full system end to end, including a Flutter consumer application, a **Next.js** War Room dashboard, a **TypeScript Express.js** API gateway, a nine-service Python orchestrator, and the underlying smart-contract layer. The platform is publicly deployed on a Sepolia-backed environment at **amttp.com**, with deployment and operational analytics supported by evidence. Holding sole technical leadership of a system of this scope demonstrates the capacity to independently architect and maintain production-grade digital infrastructure.

Technical Contribution and Security Excellence

My focus is the harder edge of compliance: privacy and verifiability. Grounded in formal cybersecurity training certified by **CompTIA Security+**, I applied defensive-architecture and threat-modelling discipline across every layer of the stack. Through **zkNAF**, I integrated **Groth16** zero-knowledge proofs so that sanctions checks and KYC confirmations may be performed without exposing personal data on a public ledger. To establish institutional-grade resilience, I subjected the contract layer to **Slither** static analysis and **Echidna** property-based fuzzing at deep iteration sequences, and complemented this with a Hardhat integration suite covering deployment, transfer logic, and policy enforcement. Together with the audited contract suite, these artefacts evidence engineering discipline.

Machine Learning, Operational Efficiency, and Intellectual Property

I designed and evaluated knowledge-distilled ML models on a dataset of **more than 625,000 Ethereum address samples**, producing distilled **XGBoost** and **LightGBM** students that retain strong fraud-detection performance while running on commodity CPUs. This eliminates the dependency on specialist GPU infrastructure and addresses major practical barrier to deploying advanced detection in regulated environments. I protected this innovation by filing **UK Patent Application No. 1026066039** on **4 March 2026**, covering the ML risk-scoring pipeline, blockchain compliance architecture, zero-knowledge identity workflows, and adaptive-friction mechanisms.

Academic Uptake and Outside Contribution

Although developed independently of any university, the research underpinning AMTTP has attracted early external recognition. Across my broader body of four research preprints, three specific papers — the **Universal Deviation Principle**, **Blind-Spot Decomposition and the Geometry of System Collapse**, and the **AMTTP architecture preprint** — were adopted in **April 2026** as recommended readings for the postgraduate course **CPE 604** by **Dr Sunday Adeola Ajagbe**, an independent academic with no prior supervisory relationship to me. Course slides and the associated module materials evidence direct integration into a live, accredited postgraduate curriculum. This adoption reflects academic interest in the underlying frameworks on their merits including fields outside financial compliance, further supporting their broader applicability.

Conclusion: Exceptional Promise

My profile is defined by **sustained independent execution as a founder**. Without a corporate brand or institutional sponsorship, I have produced a coherent body of work — a deployed platform, a public codebase, audited contracts, reproducible ML benchmarks, peer-cited research, and a UK patent filing — that demonstrates a capacity for leadership rather than mere participation. I am seeking endorsement to scale these innovations within the UK ecosystem: to **incorporate in Derby**, to **create skilled engineering and compliance roles**, to engage UK fintechs and regulators, and to advance the technical standards of British RegTech. The accompanying evidence sets out the technical, research, and engineering basis of my application.